

# Machine Learning Assignment: Customer Segmentation & Purchase Prediction

## Assignment Overview

You are a data analyst at an e-commerce company. The business team wants to understand customer behavior better to personalize marketing campaigns and predict future purchases. You have been provided with a customer transaction dataset containing customer demographics and purchase history.

Your task is to segment customers into meaningful groups and build a predictive model to forecast customer spending behavior.

## Dataset Description

You will work with a **customer transaction dataset** containing the following columns:

### Customer Demographics:

- `customer_id` : Unique identifier for each customer
- `age` : Age of the customer (years)
- `gender` : Gender (Male/Female/Other)
- `city_tier` : City classification (Tier 1, Tier 2, Tier 3)
- `membership_type` : Type of membership (Silver/Gold/Platinum)

### Purchase Behavior:

- `total_spend` : Total amount spent by customer (in currency)
- `num_transactions` : Number of transactions made
- `avg_transaction_value` : Average value per transaction
- `days_since_last_purchase` : Days since last purchase
- `num_visits` : Number of website/app visits
- `product_categories_purchased` : Number of different product categories purchased
- `discount_used` : Total discount availed (in currency)

### Target Variable:

- `high_value_customer` : Binary label (1 = High value customer, 0 = Low value customer)

[Download Dataset from here](#)

## Part 1: Data Exploration and Preprocessing (20 points)

### Tasks:

#### 1.1 Load and Inspect the Data (5 points)

- Load the dataset using `pandas`
- Display the first 10 rows
- Check the shape of the dataset
- Display data types and check for null values using `info()`
- Generate summary statistics using `describe()`

#### 1.2 Handle Missing Values (5 points)

- Identify columns with missing values
- For numerical columns, fill with the median value
- For categorical columns, fill with the mode
- Verify that no missing values remain

#### 1.3 Handle Duplicates (3 points)

- Check for duplicate rows based on `customer_id`
- Remove duplicate rows if any exist
- Report the number of duplicates removed (if any)

#### 1.4 Basic Exploratory Analysis (7 points)

- Use `value_counts()` to show the distribution of:
  - `gender`
  - `membership_type`
  - `high_value_customer`
- Create **2 visualizations** using `seaborn` or `matplotlib`:
  - Histogram showing distribution of `total_spend`
  - Box plot comparing `total_spend` across different `membership_type`
- Write 2-3 observations from your exploratory analysis

## Part 2: Customer Segmentation using Clustering (25 points)

### Tasks:

#### 2.1 Prepare Data for Clustering (7 points)

- Select numerical features for customer segmentation:
  - `age`, `total_spend`, `num_transactions`, `avg_transaction_value`, `days_since_last_purchase`, `num_visits`
- Apply Standard Scaling to normalize all selected features
- Explain in 1-2 sentences why scaling is necessary before clustering

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- For each K value, calculate the Within-Cluster Sum of Squares (WCSS)
- Create an elbow curve plot showing WCSS vs number of clusters
- State your chosen K value and reasoning (2-3 sentences)

#### 2.3 Apply K-means Clustering (10 points)

- Apply K-means clustering with your chosen optimal K value
- Add the cluster labels as a new column `customer_segment` to your original dataframe
- Display the count of customers in each cluster
- Calculate and display the mean values of key features for each cluster using `groupby`
- Write a brief profile (2-3 sentences) for each customer segment describing their characteristics

## Part 3: Predictive Modeling (35 points)

### Tasks:

#### 3.1 Feature Engineering (5 points)

- Create 2 new features from existing data. Examples:
  - `spend_per_visit = total_spend / num_visits`
  - `discount_percentage = discount_used / total_spend`
- Explain why these features might be useful (1-2 sentences)

#### 3.2 Data Preparation for Modeling (10 points)

- Separate features (X) and target variable (y = `high_value_customer`)
- Identify and apply One-Hot Encoding to categorical features ( `gender` , `city_tier` , `membership_type` )
- Split the data into training (80%) and testing (20%) sets with `random_state=42`
- Apply Standard Scaling to numerical features:
  - Important:** Fit the scaler only on training data to avoid data leakage
  - Transform both train and test sets

#### 3.3 Build Classification Model (10 points)

- Build and train ONE classification model of your choice:
  - Option 1:** Logistic Regression
  - Option 2:** Decision Tree Classifier
  - Option 3:** Random Forest Classifier
- Fit the model on training data
- Generate predictions on test data
- Briefly justify your model choice (1-2 sentences)

#### 3.4 Model Evaluation (10 points)

- Calculate and display:
  - Confusion Matrix
  - Accuracy
  - Precision
  - Recall
  - F1 Score
- Create a classification report using `classification_report()`
- Write 2-3 sentences interpreting the results (Which metric is most important for this business problem? How well is the model performing?)

## Part 4: Model Optimization and Business Insights (20 points)

### Tasks:

#### 4.1 Hyperparameter Tuning (10 points)

- Define a parameter grid with 2-3 hyperparameters for your chosen model:
  - Decision Tree:** `max_depth` , `min_samples_split`
  - Random Forest:** `n_estimators` , `max_depth`
  - Logistic Regression:** `C` , `penalty`
- Use GridSearchCV with 3-fold cross-validation ( `cv=3` )
- Report the best parameters found
- Evaluate the tuned model on test data
- Compare performance (at least accuracy and F1-score) before and after tuning in a simple table

#### 4.2 Feature Importance Analysis (5 points)

- Extract and display feature importance (for tree-based models) OR coefficients (for Logistic Regression)
- Create a horizontal bar plot showing the **top 8 most important features**
- Write 2-3 sentences interpreting what these features mean for the business

#### 4.3 Business Recommendations (5 points)

- Provide 3 specific, actionable recommendations based on your analysis:
  - 1 recommendation from clustering insights
  - 1 recommendation from model predictions
  - 1 recommendation from feature importance
- Ensure recommendations are data-driven and reference specific findings

## Submission Guidelines

Submit one file:

Jupyter Notebook (.ipynb)

Filename: `FirstName_LastName_ML_Assignment.ipynb`

Must contain:

- All cells executed with visible outputs
- Markdown explanations for all written responses
- Labeled plots with titles and axis labels
- Clean, organized code with comments

Before Submitting:

- ☒ Restart kernel and run all cells from top to bottom
- ☒ Verify all outputs are visible
- ☒ Check that all markdown cells are properly formatted
- ☒ Import all libraries at the beginning

## Mandatory Notebook Structure

```
# Machine Learning Assignment: Customer Segmentation & Purchase Prediction
Name:
Date:

## Part 1: Data Exploration and Preprocessing
### 1.1 Load and Inspect
### 1.2 Handle Missing Values
### 1.3 Handle Duplicates
### 1.4 Exploratory Analysis

## Part 2: Customer Segmentation
### 2.1 Prepare Data for Clustering
### 2.2 Determine Optimal K
### 2.3 Apply K-means and Profile Segments
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```
### 3.2 Data Preparation
### 3.3 Build Model
### 3.4 Model Evaluation

## Part 4: Optimization and Insights
### 4.1 Hyperparameter Tuning
### 4.2 Feature Importance
### 4.3 Business Recommendations
```